

all three rows are collinear; $M(H)$ singles out the ferromagnet, and only the spin splitting $\Delta(\mathbf{k}) = E_{\uparrow}(\mathbf{k}) - E_{\downarrow}(\mathbf{k})$ separates the other two

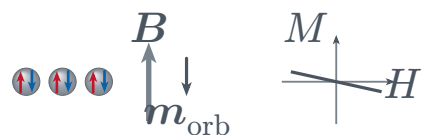
real-space
structure

response
 $M(H)$

spin-resolved
bands

spin
splitting
 $\Delta(\mathbf{k})$

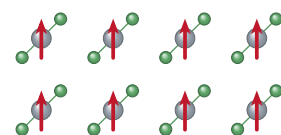
Diamagnetism



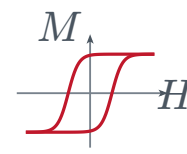
$\mathbf{m}_{\text{orb}}$ opposes $\mathbf{B}$, so
 $\chi < 0$
closed shells, $|\chi| \sim 10^{-5}$

Ferromagnetism

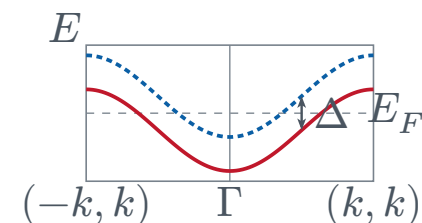
one sublattice,
every moment
parallel



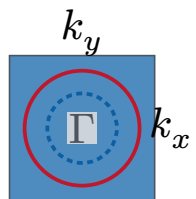
identical ligand
axes



hysteretic,
 $M_r \neq 0$



rigid s -wave
split, same sign
at every $\mathbf{k}$



concentric
contours,
majority larger

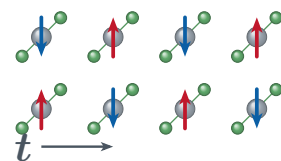
Paramagnetism



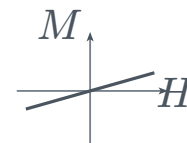
disordered moments,
 $\langle \mathbf{m} \rangle = 0$
 $\chi = C/T \sim 10^{-3} > 0$

Antiferromagnetism

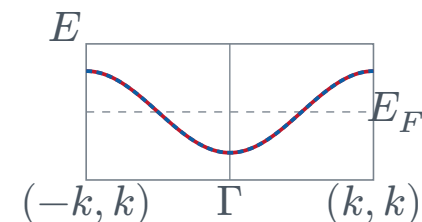
sublattices
related by a
translation $\mathbf{t}$, or
by inversion



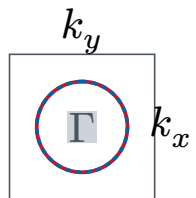
identical ligand
axes



linear, no
remanence



$E_{\uparrow}(\mathbf{k}) = E_{\downarrow}(\mathbf{k})$
everywhere



one doubly
degenerate
contour

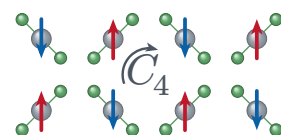
Ferrimagnetism



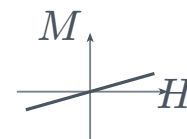
antiparallel but
 $|\mathbf{m}_A| \neq |\mathbf{m}_B|$
 $M_s = |\mathbf{M}_A + \mathbf{M}_B| \neq 0$

Altermagnetism

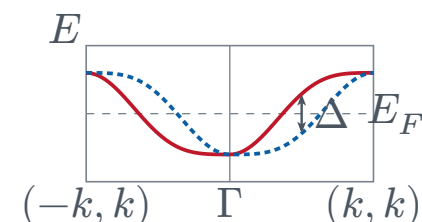
sublattices
related only by
a C_4 rotation



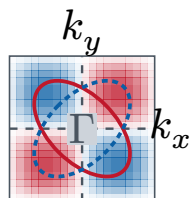
ligand axes
rotated 90°



identical to the
antiferromagnet



Δ flips between
the C_4 -related
diagonals



$\Delta \propto \sin k_x \sin k_y$,
nodal on axes

— $E_{\uparrow}$ — $E_{\downarrow}$ curves drawn on top of each other are spin degenerate ■ $\Delta > 0$ ■ $\Delta < 0$

non-relativistic limit throughout; $M(H)$ axes not to scale across panels; d -wave

altermagnet shown, g - and i -wave also exist